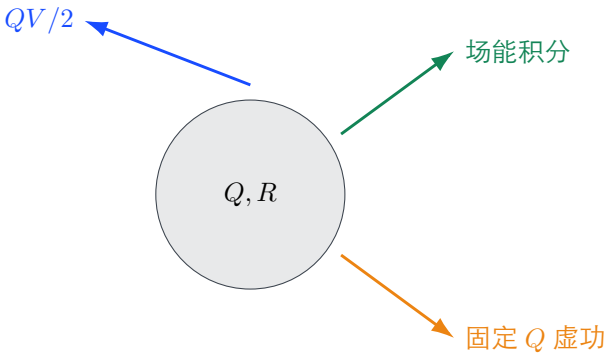
孤立带电导体球：代数、场能积分与虚功三条路径必须一致



$$U = \frac{1}{2}QV = \frac{Q^2}{8\pi\varepsilon_0 R}$$

$$U = \int_R^\infty \frac{\varepsilon_0 E^2}{2} 4\pi r^2 dr$$

$$p = -\frac{1}{4\pi R^2}\left(\frac{\partial U}{\partial R}\right)_Q = \frac{Q^2}{32\pi^2\varepsilon_0 R^4}$$